

Absolute-Value, Piecewise, and Linear-Quadratic Models — Boundaries and Open/Closed Endpoints

Every rule below has a **boundary**: the input where one formula stops and another starts. Almost every mistake on this page happens at that one input.

- A **closed** inequality ($\leq$ or $\geq$) *owns* the boundary point. A **strict** one ($<$ or $>$) does not.
- Exactly one rule may own each input. Two rules that both own it give two outputs, and neither owning it gives none.
- Read the inequality *before* you substitute. The formula that looks nearest is not always the one that applies.

Show the rule you chose, not just the number.

1. Warm up. A model is defined by

$$F(x) = \begin{cases} 2x + 1, & x < 3 \\ x^2 - 4, & x \geq 3 \end{cases}$$

Which rule owns the input $x = 1$? Name it, then evaluate $F(1)$.

Answer: _____

2. Find and fix. Using the same model

$$F(x) = \begin{cases} 2x + 1, & x < 3 \\ x^2 - 4, & x \geq 3 \end{cases}$$

Mia evaluates at the boundary and writes $F(3) = 7$. Explain in one sentence which rule Mia used and why that rule does not own $x = 3$. Then give the correct value of $F(3)$.

Answer: _____

3. Find and fix. A different model reverses which side is closed:

$$G(x) = \begin{cases} 5 - x, & x \leq -2 \\ x^2 + 1, & x > -2 \end{cases}$$

Leo has learned to “use the second rule at the boundary” and writes $G(-2) = 5$. Say what is wrong with that habit, then give the correct value of $G(-2)$.

Answer: _____

4. Find and fix. A student writes this definition:

$$H(x) = \begin{cases} x + 4, & x \leq 1 \\ 3x, & x \geq 1 \end{cases}$$

Substituting $x = 1$ into the two rules gives two different outputs, so H is not a function as written. Repair the definition by changing exactly one inequality sign so that the *top* rule keeps the boundary, and then give $H(1)$.

Answer: _____

5. A second student writes:

$$K(x) = \begin{cases} 2x, & x < 0 \\ x + 2, & x > 0 \end{cases}$$

(a) Explain why this definition fails at $x = 0$, and give two different repairs, each changing exactly one inequality sign. (*Open response — your teacher checks this part.*) (b) Repair it so that the *top* rule owns the boundary, then give $K(0)$.

Answer: _____

6. A machine part passes inspection when its measurement x , in centimetres, satisfies $|2x - 5| \leq 3$. Priya reports the passing range as $1 < x < 4$ and tells the operator that a part measuring exactly 1 does not pass. Solve $|2x - 5| = 3$ to find the two endpoints, then state whether each endpoint passes inspection and correct Priya's range.

Answer: _____

7. A ramp is modelled by

$$M(x) = \begin{cases} 4x + k, & x < 2 \\ x^2 - 1, & x \geq 2 \end{cases}$$

The builder wants no jump at the boundary: the value the top rule approaches as x climbs to 2 must equal the value the bottom rule actually takes at $x = 2$. Find the value of k that does this.

Answer: _____

8. **Challenge.** A designer wants a piecewise model built from the line $y = x - 1$ and the parabola $y = x^2 - 4x + 3$, switching from one to the other at a point where the two graphs meet, so the model has no jump. Find every point where the line and the parabola meet. Then state which of the two x -values must be written with a closed inequality if the line is the rule used first.

Answer: _____